

NPK Soil Analysis & Crop Disease-Susceptibility Report

Source file: Soil data.csv | Samples analyzed: 875

Important: N, P, K and pH values cannot by themselves diagnose a named plant disease. This report identifies nutrient-status patterns and evidence-based *susceptibility/risk* relationships. Disease diagnosis also requires crop identity, symptoms/pathogen information and environmental conditions.

Area-wise analysis (District)

Area	Samples	N range (avg)	P range (avg)	K range (avg)	pH range (avg)	Nutrient interpretation	Disease-susceptibility interpretation
Ahmedabad	9	20.0–28.9 (avg 21.9)	52.8–80.0 (avg 68.3)	65.0–80.0 (avg 76.3)	52.8–80.0 (avg 68.3)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Amreli	11	20.0–30.7 (avg 22.2)	53.6–80.0 (avg 74.1)	28.2–79.9 (avg 65.8)	53.6–80.0 (avg 74.1)	possible N deficiency	reduced vigor; greater susceptibility to some diseases
Anand	8	20.0–34.9 (avg 24.5)	78.0–80.0 (avg 79.8)	71.4–78.0 (avg 73.8)	78.0–80.0 (avg 79.8)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Ariyalur	6	20.0–20.1 (avg 20.0)	50.0–52.3 (avg 50.9)	49.5–50.0 (avg 49.8)	50.0–52.3 (avg 50.9)	possible N deficiency; possible K deficiency	reduced vigor; greater susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Arvalli	6	20.0–21.6 (avg 20.9)	67.1–77.5 (avg 73.9)	56.8–69.7 (avg 65.8)	67.1–77.5 (avg 73.9)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Bagalakote	9	25.2–48.0 (avg 33.4)	54.9–63.5 (avg 60.6)	39.7–74.5 (avg 55.9)	54.9–63.5 (avg 60.6)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Ballari	5	35.2–51.2 (avg 44.6)	45.9–57.5 (avg 54.0)	35.8–59.3 (avg 48.8)	45.9–57.5 (avg 54.0)	possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Banas Kantha	14	20.0–21.4 (avg 20.4)	54.8–80.0 (avg 74.4)	49.5–77.4 (avg 68.3)	54.8–80.0 (avg 74.4)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Bangalore Rural	4	21.2–22.9 (avg 22.2)	62.9–70.5 (avg 66.3)	42.5–53.4 (avg 49.3)	62.9–70.5 (avg 66.3)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Belagavi	15	20.0–31.7 (avg 21.8)	40.1–59.3 (avg 50.6)	29.0–72.3 (avg 48.2)	40.1–59.3 (avg 50.6)	possible P deficiency; possible K deficiency	K deficiency can increase susceptibility to several diseases; P deficiency can reduce growth/defense
Bengaluru urban	5	20.3–24.6 (avg 21.9)	47.7–73.2 (avg 63.4)	41.3–53.7 (avg 49.1)	47.7–73.2 (avg 63.4)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Bharuch	14	20.0–21.4 (avg 20.4)	54.8–80.0 (avg 74.4)	49.5–77.4 (avg 68.3)	54.8–80.0 (avg 74.4)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Bhavnagar	9	21.5–27.8 (avg 23.9)	32.2–79.2 (avg 47.8)	36.5–75.6 (avg 59.1)	32.2–79.2 (avg 47.8)	possible P deficiency	P deficiency can reduce growth/defense
Bidar	8	35.0–38.0 (avg 36.3)	47.3–48.5 (avg 47.9)	79.3–79.8 (avg 79.6)	47.3–48.5 (avg 47.9)	possible P deficiency	excess N can increase susceptibility to some diseases; P deficiency can reduce growth/defense
Botad	4	21.6–26.0 (avg 23.5)	42.1–51.0 (avg 46.7)	78.1–79.2 (avg 78.8)	42.1–51.0 (avg 46.7)	possible P deficiency	P deficiency can reduce growth/defense
Chamarajanagar	5	46.6–60.4 (avg 55.0)	62.8–71.6 (avg 66.3)	38.1–55.8 (avg 45.3)	62.8–71.6 (avg 66.3)	possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Chengalpattu	8	20.1–21.4 (avg 20.9)	40.5–55.3 (avg 47.9)	30.0–54.1 (avg 42.9)	40.5–55.3 (avg 47.9)	possible P deficiency; possible K deficiency	K deficiency can increase susceptibility to several diseases; P deficiency can reduce growth/defense
Chhotaudepur	6	26.7–35.3 (avg 31.2)	29.9–56.8 (avg 44.2)	51.3–74.0 (avg 61.8)	29.9–56.8 (avg 44.2)	possible P deficiency	excess N can increase susceptibility to some diseases; P deficiency can reduce growth/defense
Chikkaballapura	6	20.0–21.5 (avg 20.7)	71.4–76.5 (avg 73.4)	57.2–65.6 (avg 59.4)	71.4–76.5 (avg 73.4)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Chikkamagaluru	9	22.0–35.1 (avg 26.4)	58.0–78.2 (avg 70.6)	34.2–70.2 (avg 54.3)	58.0–78.2 (avg 70.6)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Chitradurga	6	21.1–22.2 (avg 21.6)	69.3–75.8 (avg 73.5)	58.2–62.0 (avg 59.7)	69.3–75.8 (avg 73.5)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone

Area	Samples	N range (avg)	P range (avg)	K range (avg)	pH range (avg)	Nutrient interpretation	Disease-susceptibility interpretation
Coimbatore	12	20.0–20.7 (avg 20.4)	36.5–49.5 (avg 42.8)	76.7–79.8 (avg 78.3)	36.5–49.5 (avg 42.8)	possible P deficiency	P deficiency can reduce growth/defense
Cuddalore	14	20.0–20.1 (avg 20.0)	62.3–77.1 (avg 70.6)	43.2–62.0 (avg 51.8)	62.3–77.1 (avg 70.6)	possible N deficiency; possible K deficiency	reduced vigor; greater susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Dakshina Kannada	9	20.1–23.1 (avg 21.1)	47.0–66.4 (avg 54.7)	29.0–55.7 (avg 40.9)	47.0–66.4 (avg 54.7)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Dang	3	34.1–44.3 (avg 39.5)	29.0–80.0 (avg 46.8)	41.3–52.0 (avg 48.3)	29.0–80.0 (avg 46.8)	possible P deficiency; possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases; P deficiency can reduce growth/defense
Davanagere	6	47.8–50.1 (avg 49.0)	74.3–75.9 (avg 75.0)	69.5–72.3 (avg 70.8)	74.3–75.9 (avg 75.0)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Devbhumi Dwarka	4	29.6–46.2 (avg 40.3)	76.0–80.0 (avg 77.4)	48.2–67.0 (avg 54.6)	76.0–80.0 (avg 77.4)	possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Dharmapuri	10	20.0–20.1 (avg 20.0)	58.2–61.5 (avg 60.2)	56.4–60.6 (avg 58.7)	58.2–61.5 (avg 60.2)	possible N deficiency	reduced vigor; greater susceptibility to some diseases
Dharwad	7	39.8–54.1 (avg 48.9)	48.2–60.1 (avg 51.3)	40.2–71.2 (avg 61.3)	48.2–60.1 (avg 51.3)	possible P deficiency	excess N can increase susceptibility to some diseases; P deficiency can reduce growth/defense
Dindigul	14	20.0–23.1 (avg 20.7)	26.9–47.0 (avg 41.3)	66.7–75.2 (avg 71.1)	26.9–47.0 (avg 41.3)	possible P deficiency	P deficiency can reduce growth/defense
Dohad	9	20.0–38.7 (avg 25.2)	44.2–78.5 (avg 68.5)	53.5–74.9 (avg 61.8)	44.2–78.5 (avg 68.5)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Erode	14	20.0–20.5 (avg 20.1)	38.1–48.1 (avg 43.8)	68.6–79.5 (avg 74.8)	38.1–48.1 (avg 43.8)	possible P deficiency	P deficiency can reduce growth/defense
Gadag	7	48.4–50.0 (avg 49.4)	51.6–53.9 (avg 52.9)	58.5–68.6 (avg 63.3)	51.6–53.9 (avg 52.9)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Gandhinagar	4	20.1–20.9 (avg 20.4)	79.4–79.7 (avg 79.6)	73.6–78.7 (avg 76.5)	79.4–79.7 (avg 79.6)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Gir Somnath	6	26.7–46.3 (avg 36.5)	47.8–52.5 (avg 50.1)	41.5–46.5 (avg 44.1)	47.8–52.5 (avg 50.1)	possible P deficiency; possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases; P deficiency can reduce growth/defense
Hassan	8	50.6–64.1 (avg 55.8)	76.8–79.9 (avg 77.8)	54.4–62.7 (avg 57.9)	76.8–79.9 (avg 77.8)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Haveri	8	27.6–49.3 (avg 41.0)	48.2–53.6 (avg 50.6)	48.8–59.6 (avg 53.5)	48.2–53.6 (avg 50.6)	possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Jamnagar	6	31.6–38.3 (avg 34.2)	69.4–74.3 (avg 71.8)	57.5–61.1 (avg 59.2)	69.4–74.3 (avg 71.8)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Junagadh	9	20.2–28.2 (avg 22.2)	62.7–77.8 (avg 72.6)	62.8–79.0 (avg 68.6)	62.7–77.8 (avg 72.6)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Kachchh	10	20.0–39.2 (avg 25.6)	42.7–74.4 (avg 62.1)	65.2–78.7 (avg 72.4)	42.7–74.4 (avg 62.1)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Kalaburagi	11	24.1–28.5 (avg 25.4)	74.6–80.0 (avg 78.8)	79.4–80.0 (avg 79.9)	74.6–80.0 (avg 78.8)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Kallakurichi	9	20.0–20.1 (avg 20.0)	56.6–67.0 (avg 59.6)	65.2–73.5 (avg 68.2)	56.6–67.0 (avg 59.6)	possible N deficiency	reduced vigor; greater susceptibility to some diseases
Kanchipuram	5	20.0–20.4 (avg 20.1)	44.6–57.6 (avg 52.0)	52.1–62.4 (avg 56.3)	44.6–57.6 (avg 52.0)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Kanniyakumari	9	20.0–21.9 (avg 20.4)	68.7–80.0 (avg 75.5)	31.0–66.4 (avg 50.3)	68.7–80.0 (avg 75.5)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Karur	8	20.0–20.1 (avg 20.0)	34.7–62.2 (avg 47.8)	56.1–75.4 (avg 67.0)	34.7–62.2 (avg 47.8)	possible N deficiency; possible P deficiency	reduced vigor; greater susceptibility to some diseases; P deficiency can reduce growth/defense
Kheda	10	20.0–22.9 (avg 20.5)	69.8–77.7 (avg 74.6)	66.0–70.2 (avg 68.3)	69.8–77.7 (avg 74.6)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Kodagu	5	47.7–51.9 (avg 49.9)	63.2–64.4 (avg 64.0)	58.2–61.0 (avg 59.8)	63.2–64.4 (avg 64.0)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases

Area	Samples	N range (avg)	P range (avg)	K range (avg)	pH range (avg)	Nutrient interpretation	Disease-susceptibility interpretation
Kolar	6	22.0–22.8 (avg 22.2)	57.2–73.9 (avg 69.6)	41.2–51.4 (avg 46.6)	57.2–73.9 (avg 69.6)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Koppal	7	49.3–67.0 (avg 56.5)	75.3–79.4 (avg 77.7)	55.4–75.3 (avg 65.4)	75.3–79.4 (avg 77.7)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Krishnagiri	10	20.0–20.0 (avg 20.0)	34.5–65.1 (avg 53.8)	40.5–56.5 (avg 49.5)	34.5–65.1 (avg 53.8)	possible N deficiency; possible K deficiency	reduced vigor; greater susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Madurai	13	20.0–20.1 (avg 20.1)	36.6–74.6 (avg 57.2)	60.8–75.7 (avg 67.4)	36.6–74.6 (avg 57.2)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Mahesana	10	20.0–22.4 (avg 20.6)	43.4–70.0 (avg 60.5)	36.4–59.0 (avg 50.6)	43.4–70.0 (avg 60.5)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Mahisagar	6	20.0–25.2 (avg 22.5)	49.8–73.2 (avg 58.0)	69.1–78.4 (avg 74.4)	49.8–73.2 (avg 58.0)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Mandya	7	20.1–34.1 (avg 25.5)	58.2–74.0 (avg 69.7)	47.6–62.0 (avg 56.0)	58.2–74.0 (avg 69.7)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Mayiladuthurai	5	20.0–21.9 (avg 20.4)	49.3–68.6 (avg 59.5)	51.2–69.2 (avg 59.1)	49.3–68.6 (avg 59.5)	possible N deficiency	reduced vigor; greater susceptibility to some diseases
Morbi	5	29.7–38.5 (avg 35.8)	77.3–79.2 (avg 78.0)	62.6–64.5 (avg 63.6)	77.3–79.2 (avg 78.0)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Mysuru	9	20.0–20.5 (avg 20.2)	64.0–76.3 (avg 73.2)	50.6–75.3 (avg 59.4)	64.0–76.3 (avg 73.2)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Nagapattinam	6	20.0–20.2 (avg 20.1)	54.5–70.8 (avg 63.8)	51.4–59.9 (avg 56.3)	54.5–70.8 (avg 63.8)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Namakkal	15	20.0–21.9 (avg 20.2)	47.6–72.0 (avg 57.8)	65.7–79.0 (avg 74.0)	47.6–72.0 (avg 57.8)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Narmada	5	22.9–35.0 (avg 32.0)	24.3–78.3 (avg 63.2)	36.9–71.4 (avg 62.7)	24.3–78.3 (avg 63.2)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Navsari	6	26.0–35.2 (avg 31.3)	79.9–80.0 (avg 80.0)	71.4–78.5 (avg 76.0)	79.9–80.0 (avg 80.0)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Panch Mahals	7	29.3–39.1 (avg 34.0)	48.1–73.5 (avg 63.8)	45.8–72.7 (avg 61.4)	48.1–73.5 (avg 63.8)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Patan	9	20.0–20.2 (avg 20.1)	58.9–73.3 (avg 67.3)	72.8–79.8 (avg 77.6)	58.9–73.3 (avg 67.3)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Perambalur	4	20.0–20.0 (avg 20.0)	34.6–43.0 (avg 39.4)	62.3–66.1 (avg 64.3)	34.6–43.0 (avg 39.4)	possible N deficiency; possible P deficiency	reduced vigor; greater susceptibility to some diseases; P deficiency can reduce growth/defense
Porbandar	3	21.6–33.1 (avg 25.5)	36.4–76.8 (avg 56.9)	75.6–80.0 (avg 78.5)	36.4–76.8 (avg 56.9)	possible P deficiency	P deficiency can reduce growth/defense
Pudukkottai	13	20.0–20.1 (avg 20.1)	43.3–58.4 (avg 50.0)	46.5–57.4 (avg 52.8)	43.3–58.4 (avg 50.0)	possible P deficiency	P deficiency can reduce growth/defense
Raichur	7	20.0–20.1 (avg 20.1)	44.7–51.6 (avg 49.8)	47.3–49.2 (avg 48.2)	44.7–51.6 (avg 49.8)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Rajkot	11	20.1–35.4 (avg 24.2)	76.1–80.0 (avg 79.2)	71.6–78.0 (avg 73.7)	76.1–80.0 (avg 79.2)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Ramanathapuram	11	20.0–20.1 (avg 20.1)	47.7–66.2 (avg 55.1)	57.1–79.9 (avg 74.7)	47.7–66.2 (avg 55.1)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Ramangara	5	42.5–46.4 (avg 44.4)	66.3–77.2 (avg 70.0)	43.8–50.4 (avg 47.3)	66.3–77.2 (avg 70.0)	possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Ranipet	7	20.0–20.1 (avg 20.0)	63.0–65.5 (avg 63.9)	40.1–43.6 (avg 42.4)	63.0–65.5 (avg 63.9)	possible N deficiency; possible K deficiency	reduced vigor; greater susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Sabar Kantha	8	20.0–20.2 (avg 20.1)	74.6–77.9 (avg 76.6)	58.9–66.3 (avg 64.2)	74.6–77.9 (avg 76.6)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Salem	20	20.0–21.3 (avg 20.1)	37.6–54.6 (avg 46.1)	56.0–74.4 (avg 66.5)	37.6–54.6 (avg 46.1)	possible P deficiency	P deficiency can reduce growth/defense
Shivamogga	7	24.8–32.7 (avg 27.5)	42.7–70.3 (avg 57.0)	53.3–70.4 (avg 61.6)	42.7–70.3 (avg 57.0)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Sivaganga	12	20.0–20.8 (avg 20.1)	32.1–69.5 (avg 50.9)	57.9–71.3 (avg 66.0)	32.1–69.5 (avg 50.9)	possible N deficiency; possible P deficiency	reduced vigor; greater susceptibility to some diseases; P deficiency can reduce growth/defense
Surat	9	20.0–27.2 (avg 22.5)	64.6–77.3 (avg 74.0)	59.6–72.4 (avg 64.6)	64.6–77.3 (avg 74.0)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone

Area	Samples	N range (avg)	P range (avg)	K range (avg)	pH range (avg)	Nutrient interpretation	Disease-susceptibility interpretation
Surendranagar	10	22.4–48.2 (avg 30.8)	67.4–75.7 (avg 72.7)	59.8–79.9 (avg 70.9)	67.4–75.7 (avg 72.7)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Tapi	7	21.0–30.2 (avg 25.1)	74.0–77.2 (avg 75.5)	67.9–77.9 (avg 71.0)	74.0–77.2 (avg 75.5)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Tenkasi	10	20.0–20.1 (avg 20.0)	71.4–78.7 (avg 76.0)	59.9–79.7 (avg 74.2)	71.4–78.7 (avg 76.0)	possible N deficiency	reduced vigor; greater susceptibility to some diseases
Thanjavur	14	20.0–21.3 (avg 20.3)	31.5–62.7 (avg 46.6)	28.6–74.2 (avg 53.3)	31.5–62.7 (avg 46.6)	possible N deficiency; possible P deficiency; possible K deficiency	reduced vigor; greater susceptibility to some diseases; K deficiency can increase susceptibility to several diseases; P deficiency can reduce growth/defense
The Nilgiris	4	20.6–36.0 (avg 27.4)	78.0–79.7 (avg 79.0)	46.8–78.3 (avg 63.3)	78.0–79.7 (avg 79.0)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Theni	8	20.0–20.4 (avg 20.2)	50.0–66.6 (avg 59.5)	68.5–77.5 (avg 73.6)	50.0–66.6 (avg 59.5)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Thiruvallur	14	20.0–20.2 (avg 20.1)	44.8–80.0 (avg 62.2)	28.6–50.0 (avg 38.8)	44.8–80.0 (avg 62.2)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Thiruvaurur	10	20.0–20.4 (avg 20.0)	50.8–77.6 (avg 65.4)	31.3–79.9 (avg 61.4)	50.8–77.6 (avg 65.4)	possible N deficiency	reduced vigor; greater susceptibility to some diseases
Tiruchippalli	14	20.0–22.8 (avg 20.2)	38.0–77.8 (avg 56.8)	64.7–78.9 (avg 69.9)	38.0–77.8 (avg 56.8)	possible N deficiency; possible P deficiency	reduced vigor; greater susceptibility to some diseases; P deficiency can reduce growth/defense
Tirunelveli	9	20.2–27.7 (avg 22.4)	29.4–72.1 (avg 56.3)	33.5–69.3 (avg 58.5)	29.4–72.1 (avg 56.3)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Tirupathur	6	20.0–20.3 (avg 20.1)	58.3–66.6 (avg 63.4)	51.2–61.6 (avg 58.0)	58.3–66.6 (avg 63.4)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Tiruppur	13	20.0–20.1 (avg 20.1)	40.5–48.9 (avg 45.2)	77.8–79.8 (avg 79.0)	40.5–48.9 (avg 45.2)	possible P deficiency	P deficiency can reduce growth/defense
Tiruvannamalai	18	20.0–22.7 (avg 20.4)	45.3–64.8 (avg 55.8)	33.3–57.3 (avg 44.9)	45.3–64.8 (avg 55.8)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Tumakuru	10	22.5–28.5 (avg 25.5)	62.6–74.3 (avg 68.0)	51.1–62.0 (avg 56.4)	62.6–74.3 (avg 68.0)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Tuticorn	12	20.0–20.0 (avg 20.0)	45.4–71.0 (avg 56.2)	34.3–80.0 (avg 69.9)	45.4–71.0 (avg 56.2)	possible N deficiency	reduced vigor; greater susceptibility to some diseases
Udupi	7	20.0–40.0 (avg 31.6)	38.0–72.3 (avg 58.2)	26.0–51.6 (avg 43.5)	38.0–72.3 (avg 58.2)	possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases
Uttara Kannada	12	20.0–31.9 (avg 25.2)	21.9–68.2 (avg 53.1)	26.4–70.9 (avg 44.7)	21.9–68.2 (avg 53.1)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Vadodara	8	20.0–40.4 (avg 26.3)	55.4–73.2 (avg 63.5)	22.9–78.8 (avg 58.3)	55.4–73.2 (avg 63.5)	possible K deficiency	K deficiency can increase susceptibility to several diseases
Valsad	6	29.7–34.0 (avg 31.6)	75.6–76.7 (avg 76.1)	60.4–67.2 (avg 63.7)	75.6–76.7 (avg 76.1)	no dominant low N/P/K range	excess N can increase susceptibility to some diseases
Vellore	7	20.0–20.2 (avg 20.1)	34.2–43.8 (avg 37.9)	48.7–58.4 (avg 54.7)	34.2–43.8 (avg 37.9)	possible P deficiency	P deficiency can reduce growth/defense
Vijayanagar	6	20.0–42.6 (avg 32.5)	37.1–71.0 (avg 53.1)	31.5–54.8 (avg 39.8)	37.1–71.0 (avg 53.1)	possible P deficiency; possible K deficiency	excess N can increase susceptibility to some diseases; K deficiency can increase susceptibility to several diseases; P deficiency can reduce growth/defense
Vijayapura	13	20.0–25.7 (avg 20.9)	68.9–79.2 (avg 75.8)	53.7–70.2 (avg 61.1)	68.9–79.2 (avg 75.8)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Villupuram	13	20.0–20.1 (avg 20.0)	44.3–78.2 (avg 58.2)	47.2–67.3 (avg 56.1)	44.3–78.2 (avg 58.2)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Virudhunagar	11	20.0–20.3 (avg 20.1)	20.0–62.6 (avg 43.0)	46.9–77.6 (avg 67.1)	20.0–62.6 (avg 43.0)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone
Yadgir	6	22.1–28.7 (avg 25.0)	62.8–75.8 (avg 70.1)	75.0–78.6 (avg 76.4)	62.8–75.8 (avg 70.1)	no dominant low N/P/K range	no specific disease risk inferred from NPK alone

How to interpret N, P and K for disease prediction

- **Nitrogen (N):** deficiency commonly causes poor growth and yellowing. Nutrient-stressed plants can be more susceptible to infection; excessive N can also increase susceptibility to some diseases depending on the crop/pathogen.
- **Phosphorus (P):** deficiency causes stunting and dark-green/purple foliage. Its effect on disease varies by crop and pathogen; therefore P alone should not be mapped to one disease.
- **Potassium (K):** adequate K generally supports plant strength and defense. K deficiency can increase susceptibility to several diseases, while the effect of extra K varies by disease and crop.
- **pH:** pH changes nutrient availability and therefore can indirectly affect plant health. A crop-specific pH target is needed before classifying a sample as deficient or excessive.

Evidence basis

University and plant-pathology sources emphasize that nutrient deficiencies can increase susceptibility to infection and that nutrient–disease relationships are crop/pathogen specific. For example, potassium has been associated with reduced severity of wheat leaf rust/leaf blight and rice sheath blight when adequate, while nitrogen and phosphorus can have different effects depending on the disease.

Method used in this report

Because the CSV does not establish a universal laboratory extraction method, units, crop-specific sufficiency thresholds, or measured disease labels, N/P/K status was summarized using the dataset's own lower 25%, middle 50%, and upper 25% ranges. This avoids inventing agronomic cutoff values. These ranges are suitable for exploratory ML feature engineering, but should not be presented as clinical/agronomic deficiency thresholds.